

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 1 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Introduction to Control Systems

A **control system** is an interconnection of physical components arranged to *regulate, direct, or command* the behaviour of another system so as to obtain a desired response. The fundamental goal is to **maintain a specified output** in spite of disturbances or parameter variations.

Formal Definition: A control system is a system in which the output quantity is controlled by varying the input quantity. It comprises a *controller*, a *plant* (process), and — in feedback systems — a *sensor* that measures the output.

1.1 Key Terminology

Term	Definition
Plant / Process	The physical object or process to be controlled (e.g. motor, furnace, tank).
Controlled Variable	The output quantity measured and maintained at a desired level (e.g. speed, temperature).
Reference Input $R(s)$	The desired (set-point) value of the controlled variable, supplied externally.
Error Signal $E(s)$	$E = R - B$; difference between reference input and feedback signal. Drives the controller.
Actuator	Device (motor, valve, heater) that physically drives the plant based on controller output.
Sensor / Transducer	Measures the actual output and converts it into a form comparable to the reference signal.
Feedback Signal $B(s)$	Signal derived from the output (via sensor) fed back to the summing junction.
Disturbance	Unwanted external or internal signal that adversely affects the plant output.
Controller	Element that processes the error signal and generates corrective action $U(s)$.

1.2 Industrial Control Examples

- Furnace Temperature Control:** Thermostat measures temperature, compares with set-point, switches heater on/off — classic *closed loop on-off* controller.
- DC Motor Speed Control:** Tachometer feeds back actual RPM; controller adjusts armature voltage to maintain desired speed despite load variations.
- Liquid Level Control:** Float/electronic sensor detects tank level; valve opened/closed to keep level at set value.

4. **Missile Guidance System:** Inertial/radar sensor tracks actual position and continuously corrects trajectory — high-speed closed loop system.
5. **Aircraft Auto-Pilot:** Gyroscopes and accelerometers measure altitude, heading, speed; auto-pilot adjusts control surfaces to match reference values.

2. Transfer Function

The transfer function is the primary mathematical tool to model, analyse, and design LTI control systems. It converts the time-domain differential equation into an algebraic expression in the s -domain (Laplace domain).

Definition: The **Transfer Function** $G(s)$ of a linear time-invariant (LTI) system is the ratio of the Laplace transform of the *output* to the Laplace transform of the *input*, with all initial conditions assumed **zero**.

$$G(s) = \left. \frac{\mathcal{L}\{\text{output}\}}{\mathcal{L}\{\text{input}\}} \right|_{\text{zero I.C.}} = \frac{Y(s)}{U(s)}$$

2.1 Poles, Zeros, and Order

The denominator polynomial set to zero gives the **characteristic equation**. Its roots are the **poles** — they determine stability and transient response. Roots of the numerator are the **zeros**. The **order** is the highest power of s in the denominator.

2.2 Properties of Transfer Function

Properties of Transfer Function	
1	Defined and valid only for LTI (Linear, Time-Invariant) systems.
2	Independent of input magnitude — depends only on system parameters (R, C, L, K , etc.).
3	Setting denominator = 0 gives the characteristic equation ; its roots are the poles .
4	Setting numerator = 0 gives the zeros of the system.
5	Order of system = degree of the denominator polynomial (n).
6	All initial conditions assumed zero — it is an <i>input-output</i> model only.
7	Does not describe internal (state) variables of the system.

2.3 Standard Form

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}, \quad n \geq m \text{ (proper system)}$$

2.4 Example — Series RC Circuit

Given: Series RC circuit. Input = $V_i(s)$, Output = $V_o(s)$ (voltage across C).
Applying voltage divider in s -domain:

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau}, \quad \tau = RC$$

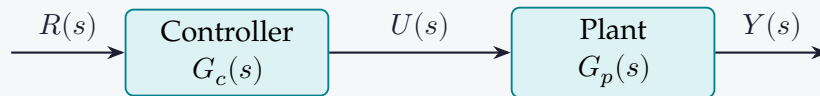
Analysis: First-order system ($n = 1$). One pole at $s = -1/\tau$, no finite zeros. Time constant $\tau = RC$ determines charging speed.

3. Open Loop Control System

In an open loop system, the controller generates a control signal based *only* on the reference input. The actual output is **never measured or compared** — the system cannot self-correct.

Definition: An **open loop control system** is one in which the control action is *independent of the output*. The output has no effect on the control signal.

Open Loop Block Diagram



$$\frac{Y(s)}{R(s)} = G_c(s) \cdot G_p(s)$$

3.1 Characteristics of Open Loop Systems

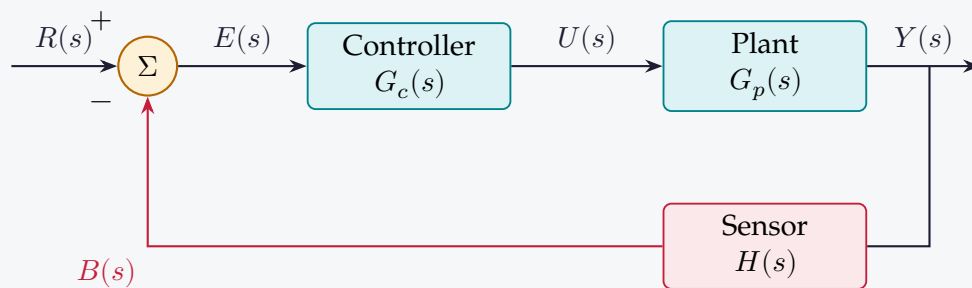
Feature	Open Loop Behaviour
Accuracy	Low — errors from disturbances or parameter changes are not corrected.
Stability	Generally stable; no feedback loop to introduce phase shift or oscillation.
Complexity	Simple design; fewer components; easy to maintain.
Cost	Lower — no sensors, feedback elements, or comparators needed.
Disturbance	Output permanently affected; no self-correction mechanism.
Calibration	Must be re-calibrated frequently to maintain accuracy.
Real Examples	Washing machine (timed cycle), traffic signal timer, electric toaster, stepper motor.

4. Closed Loop (Feedback) Control System

A closed loop system **continuously monitors the output** and uses the measured value to correct the control action. The difference between reference and actual output is the **error**, which drives the controller.

Definition: A closed loop (feedback) control system is one in which the output is measured, compared with the reference input, and the resulting *error signal* is used to reduce the deviation. Feedback is typically **negative**.

Closed Loop Block Diagram (Negative Feedback)



4.1 Closed Loop Transfer Function (CLTF)

Derivation (Unity Feedback, $H(s) = 1$):

$$\begin{aligned} E(s) &= R(s) - Y(s) \\ Y(s) &= G(s) \cdot E(s) = G(s)[R(s) - Y(s)] \\ Y(s)[1 + G(s)] &= G(s) \cdot R(s) \end{aligned}$$

Unity Feedback ($H(s) = 1$):

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

Non-Unity Feedback ($H(s) \neq 1$):

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}, \quad G(s) = G_c(s) \cdot G_p(s)$$

$G(s)H(s)$ = **open loop transfer function (OLTF)**. $1 + G(s)H(s) = 0$ = **characteristic equation**.

4.2 Open Loop vs. Closed Loop — Comparison

Parameter	Open Loop	Closed Loop
Feedback	Absent	Present (negative)
Accuracy	Lower	Higher
Stability	More stable	May become unstable
Disturbance effect	High; no correction	Reduced; self-correcting
Bandwidth	Narrow	Wider
Sensitivity	High to parameter changes	Reduced (robust)
Cost & Complexity	Low	Higher
Real Examples	Toaster, timer, stepper motor	Thermostat, servo drive, auto-pilot

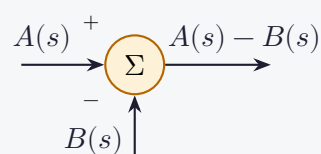
5. Block Diagram Reduction

A **block diagram** is a pictorial representation of a control system. For complex multi-loop systems, **algebraic reduction rules** simplify the diagram step-by-step until a single overall TF $Y(s)/R(s)$ is obtained.

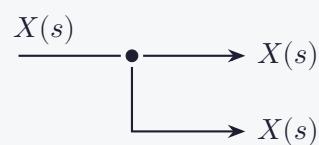
Summing junction: Circle with $+/−$ signs; algebraically adds/subtracts incoming signals.

Takeoff point: Node from which the same signal is distributed to multiple branches unchanged.

Summing Junction and Takeoff Point



Summing Junction



Takeoff Point

5.1 Reduction Rules

Operation	Description	Equivalent TF
Blocks in Series	Output of G_1 feeds input of G_2 ; same signal chain	$G_1(s) \cdot G_2(s)$
Blocks in Parallel	Same input, individual outputs summed at a junction	$G_1(s) + G_2(s)$
Feedback Loop	Forward path G , feedback path H (negative feedback)	$\frac{G}{1 + GH}$
Summing pt. forward	Shift summing junction past block G to the right	Divide shifted branch by G
Summing pt. backward	Shift summing junction past block G to the left	Multiply shifted branch by G
Takeoff pt. forward	Shift takeoff past block G to the right	Multiply branch by G
Takeoff pt. backward	Shift takeoff past block G to the left	Divide branch by G

5.2 Reduction Procedure (Step-by-Step)

- Step 1:** Identify and combine all **cascade (series)** blocks into a single block.
- Step 2:** Identify and combine all **parallel** paths into a single block.
- Step 3:** Eliminate all **feedback loops** using $G/(1 \pm GH)$.
- Step 4:** If needed, **shift** summing junctions or takeoff points to untangle crossed paths.
- Step 5:** Repeat steps 1–4 until only a **single block** $Y(s)/R(s)$ remains.

Exam Tip

Always maintain the correct sign (+/−) at every summing junction while shifting. Careless sign errors are the most common mistake in block diagram problems.

6. Signal Flow Graph (SFG) Analysis

A Signal Flow Graph is an alternative graphical method to represent and solve linear algebraic equations describing a control system. Unlike block diagrams — which require step-by-step reduction — an SFG is solved *directly* using **Mason's Gain Formula** in a single pass.

Definition: A **Signal Flow Graph (SFG)** is a directed graph of *nodes* (signals/variables) connected by *directed branches* (carrying gain/transmittance values). It is a compact representation of the system's simultaneous equations.

6.1 Basic Terminology

Term	Meaning / Definition
Node	Represents a signal or variable in the system (shown as a small circle).
Branch	Directed arrow between two nodes; labelled with the <i>transmittance</i> (gain).
Source Node	Input node — has only outgoing branches; no incoming branches.
Sink Node	Output node — has only incoming branches; no outgoing branches.
Mixed Node	A node with both incoming and outgoing branches (intermediate variable).
Forward Path	Continuous directed path from source to sink in which no node is visited more than once .
Forward Path Gain P_k	Product of all branch gains along the k -th forward path.
Loop	Closed directed path beginning and ending at the same node, no node repeated .
Loop Gain L_i	Product of all branch gains around the i -th loop.
Non-Touching Loops	Two or more loops that do not share any common node .

6.2 Mason's Gain Formula

$$T = \frac{Y(s)}{R(s)} = \frac{\sum_{k=1}^N P_k \Delta_k}{\Delta}$$

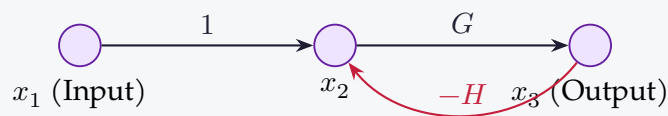
Symbol	Definition
N	Total number of forward paths from source to sink.
P_k	Gain of the k -th forward path (product of branch gains along that path).
Δ	Graph determinant = $1 - \sum_i L_i + \sum_{i,j} L_i L_j - \sum_{i,j,k} L_i L_j L_k + \dots$
$\sum L_i$	Sum of gains of all individual loops in the graph.
$\sum L_i L_j$	Sum of gain-products of all pairs of non-touching loops .
$\sum L_i L_j L_k$	Sum of gain-products of all triplets of mutually non-touching loops .
Δ_k	Cofactor of the k -th path: value of Δ after removing all loops that <i>touch</i> forward path k .

6.3 Steps to Apply Mason's Formula

- 1. Identify forward paths:** Trace every path from source to sink (no repeated nodes). Note gain P_k of each.
- 2. Identify all loops:** Find every closed path. Compute loop gain L_i for each.
- 3. Find non-touching loop combinations:** Identify all pairs, triplets, etc. sharing no common node.
- 4. Compute graph determinant:** $\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$
- 5. Compute cofactor Δ_k :** For each forward path k , remove all loops touching that path from Δ .
- 6. Apply Mason's formula:** $T = \frac{\sum P_k \Delta_k}{\Delta}$.

6.4 Worked Example — Simple Feedback System

SFG Representation



Applying Mason's formula:

- Forward path: $P_1 = 1 \times G = G$
- Loop: $L_1 = G \times (-H) = -GH$
- No non-touching loop pairs (only one loop).
- $\Delta = 1 - L_1 = 1 - (-GH) = 1 + GH$
- $\Delta_1 = 1$ (loop L_1 touches forward path P_1 , so removed from Δ)

$$T = \frac{P_1 \Delta_1}{\Delta} = \frac{G \cdot 1}{1 + GH} = \frac{G}{1 + GH} \quad \checkmark$$

Matches the CLTF from block diagram reduction — both methods are equivalent.

6.5 SFG vs. Block Diagram — Comparison

Aspect	Block Diagram	Signal Flow Graph
Representation	Blocks, summing junctions, takeoff points	Directed nodes and weighted branches
Reduction Method	Step-by-step algebra; multiple passes	Mason's formula — single direct computation
Complex Systems	Tedious; can become cluttered and error-prone	Compact, systematic, less error-prone
Gain Formula	No direct formula; reduce iteratively	$T = \sum P_k \Delta_k / \Delta$
Preferred For	Simple systems; visual understanding	Multi-loop systems; quick TF computation

Quick Revision — Module 1

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Key Formulas at a Glance

Transfer Function	$G(s) = Y(s)/U(s)$ (zero initial conditions)
CLTF (Unity feedback)	$T(s) = G(s)/[1 + G(s)]$
CLTF (Non-unity)	$T(s) = G(s)/[1 + G(s)H(s)]$
Characteristic Equation	$1 + G(s)H(s) = 0$
Series blocks	$G_{eq} = G_1 \cdot G_2$
Parallel blocks	$G_{eq} = G_1 + G_2$
Feedback reduction	$G_{eq} = G/(1 \pm GH)$
Mason's Gain Formula	$T = \frac{\sum_k P_k \Delta_k}{\Delta}$
Graph Determinant	$\Delta = 1 - \sum L_i + \sum L_i L_j - \dots$

One-Line Definitions

- **Control System:** Interconnection of components to regulate output to a desired value.
- **Transfer Function:** Ratio of Laplace output to Laplace input at zero initial conditions.
- **Poles:** Roots of denominator of $G(s)$ — determine stability and transient response.
- **Zeros:** Roots of numerator of $G(s)$ — affect frequency response shape.
- **Open Loop:** Control action independent of output; no self-correction.
- **Closed Loop:** Output fed back and compared with reference; error drives controller.
- **Error Signal:** $E(s) = R(s) - B(s)$; difference between reference and feedback.
- **SFG:** Directed graph of nodes and branches representing system equations.
- **Forward Path:** Path from source to sink with no repeated nodes.
- **Loop Gain:** Product of all branch gains around a closed loop.
- **Cofactor Δ_k :** Graph determinant Δ with all loops touching path k removed.

Frequently Asked / Viva Points

Q1. Why are initial conditions zero in TF?

TF is an input-output model; it characterises the system, not a specific response.

Frequently Asked / Viva Questions (contd.)

Zero I.C. ensures the TF is a property of the system alone.

Q2. What is the order of a system?

Highest power of s in the denominator of $G(s)$.

Q3. Difference between open loop and closed loop?

Open loop: no feedback, no self-correction. Closed loop: feedback present, error-driven, self-correcting.

Q4. What is the characteristic equation?

$1 + G(s)H(s) = 0$. Its roots (poles of CLTF) determine stability.

Q5. When is a system proper?

When degree of denominator $\geq$ degree of numerator ($n \geq m$).

Q6. What is a takeoff point?

A node from which the same signal branches to multiple paths without change.

Q7. What is Δ_k in Mason's formula?

Cofactor of the k -th forward path — Δ computed after removing all loops that touch path k .

Q8. Can SFG and block diagram give different results?

No — both are equivalent representations; Mason's formula gives the same TF as block diagram reduction.

Q9. What are non-touching loops?

Loops that share no common node. Their gain products appear in Δ .

Q10. Why is negative feedback preferred?

It reduces error, improves accuracy, reduces sensitivity to parameter variations, and widens bandwidth.

Comparison Snapshot

Property	Open Loop	Closed Loop
Feedback	No	Yes (negative)
Accuracy	Low	High
Stability	More stable	Can oscillate
Cost	Low	High
Disturbance	Not corrected	Self-corrected